The first split in an anchored string action

Human Compression project: a component of the L019 recovery

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a transitive permutation group on n points, generated by nonidentity involutions satisfying the string relations. Assume every maximal omission $G_i = \langle \rho_j : j \neq i \rangle$ is intransitive. An edge of colour i is eligible if its endpoints lie in different G_i-orbits. An endpoint anchor e is moved by ρ_0 and fixed by every later mark.

We acquire the paired-cut bound, the unique-crossing lemma, the actual head-support proof and the ambient interval deficit. These are statements about the full permutation action; they require neither parity nor the intersection property.

Lemma 1 (The anchor determines one side)**.** *Suppose an endpoint anchor e exists. The noncommutation diagram is connected. Suppose a unique eligible edge of colour $j > 0$ has omission orbits A, B, with $e \in A$. Then $G_{<j}$ is transitive on A and $G_{>j}$ fixes A pointwise. If this split is not perfect, both sides are nontrivial and some mark of index less than j acts on both sides.*

Proof. The component of the noncommutation diagram containing the first mark generates a group H. Every other component centralizes H and fixes e. Thus the full orbit of e is its H-orbit, so H is transitive. A permutation centralizing a transitive group and fixing one point fixes all points. Hence there is no other nontrivial component.

Put $L = G_{<j}$ and $R = G_{>j}$. These groups commute, R fixes e, and $LR = G_j$ is transitive on A. Consequently $e^L = A$, and centralization makes R fix A pointwise. The side A is nontrivial since ρ_0 moves e. If B were a singleton, all of L would fix it and the split would be perfect. Otherwise nonperfection means that some left mark acts on B. To find a shared mark, if no left mark were shared then each would act on precisely one side. Their side could not change at adjacent positions, by the connected noncommutation diagram. The first left mark moves $e \in A$, so all would be supported on A, a contradiction. □

Lemma 2 (The head bound in its exact rank range)**.** *In the setting of the acquired head-support proof, suppose a unique cut is at i, its sides are nontrivial, the group $G_{<i}$ fixes its second side, and some later mark is shared. If $2r > n$, the final shared index h satisfies $h < r - 1$. For the support X of $G_{>h}$, the same proof gives*

$$2h \leq n - |X| - 1. \tag{1}$$

At $i > 0$, the tail is supported on the second side. At $i = 0$, call the side containing that tail the second side.

Proof. The first five lemmas of the acquired head-support manuscript have no global rank hypothesis. Its last-mark lemma says $n \geq 2r$ if the final mark is shared, or if $i > 0$ and it acts on the first side. Both alternatives contradict the present hypothesis. Connectedness forces all marks after h onto the same side: two adjacent nonidentity marks with disjoint supports would commute. At $i = 0$ the side names may be interchanged. This is exactly the placement used in the acquired proof.

The remaining fixed-point count there is unchanged. At $i = 0$, its two disjoint path pieces have respectively $h + 1$ and h fixed points. At $i > 0$, its transitive-side case gives $|A| + 1 \geq 2i + 3 = 2h + 1$; its intransitive-side case gives $|A| + (h - i) \geq 2h + 1$. These inequalities use only the commuting-orbit, internal-pair and path lemmas, whose full weak equality cases are retained. Thus the tail fixes at least $2h+1$ points, proving (1). No step of this oriented argument uses the stronger rank assumption needed elsewhere to obtain its orientation. □

Theorem 3. *Suppose an endpoint anchor exists and $2r > n$. There is a colour with a unique eligible edge. The least such colour j is a perfect split. If A is the side containing e, then*

$$2j \leq |A| - 1. \tag{2}$$

Proof. Full transitivity ensures at least one eligible edge of each colour. If all had at least two, the acquired paired-cut bound would give $2r \leq n$. Thus j exists. If $j = 0$, the anchor is a singleton G_0-orbit. The unique-crossing lemma gives exactly one other orbit, so the split is perfect and (2) is equality.

Let $j > 0$, and suppose this split is not perfect. Use Lemma 1 and reverse all labels. The crossing index is $i' = r - 1 - j$; its first side is the old B, and its second side is the old A. The earlier group in this reversed order is the old $G_{>j}$, which fixes A. Lemma 2 applies. If g is the first shared index in the original order, the reversed final shared index is $h' = r - 1 - g$. Thus $0 < g < j$, and, with $X = \mathrm{supp}(G_{<g})$,

$$2(r - 1 - g) \leq n - |X| - 1. \tag{3}$$

For $j < r - 1$ the support placement is already fixed by Lemma 2. If $j = r - 1$, its reversed crossing is zero. The final reversed tail contains the original ρ_0, which moves $e \in A$, so that tail is on A. This checks the endpoint naming as well.

Every colour in the nonempty prefix $\{0, \ldots, g-1\}$ has at least two ambient eligible edges, by minimality of j. Its group is intransitive, being contained in G_j, and has at least one nontrivial orbit. The acquired ambient interval deficit therefore gives

$$2g \leq |X| - 1. \tag{4}$$

Adding (3) and (4) yields $2r - 2 \leq n - 2$, a contradiction. This proves perfection.

If $j > 0$, apply the same interval deficit to the full prefix $\{0, \ldots, j-1\}$. At a perfect split it is transitive on A, fixes B, and therefore has precisely one nontrivial orbit and support A. Its j colours are all paired by minimality, giving (2). □

Attribution and scope. The fracture and split framework is due to the cited Cameron–Fernandes–Leemans sources and their earlier work with Mixer. The complete acquired orbit-cut, path and interval proofs retain those attributions and their separately disclosed contributions. The present deduction uses the endpoint anchor to fix the orientation, then pays the full interval orbit deficit at the first unique cut. It replaces the longer proposed two-anchor density route for an anchored parent, without needing a finite exceptional list or a quotient intersection property.

For a critical even-degree symmetric factor of rank $m/2+1$, and for its anchored odd-degree tail of rank $(d+1)/2$, the theorem supplies a perfect split and (2). It does not force that split to be interior, make its internal factor trivial, or pay its leaf kernels. In particular $j = 0$ is allowed. The general sufficient-split and stable correspondence obligations remain separate. No new computation, whole original, conjecture, Lean or measurement credit is claimed.